
VANGUARD-32

Autonomous Smart Perimeter Monitoring & Warning Platform

ATmega32-Based Embedded Systems Design
Project Proposal

1. Motivation

Problem Description & Importance

Problem Description

- Conventional perimeter systems often rely on a single trigger source such as PIR or optical tripwire sensing.
- False alarms can occur because of wind, foliage, small animals, and environmental disturbances.
- Passive systems cannot aim toward the triggered sector or locally verify whether a warm object is present.

Why It Matters

- Remote sites, industrial zones, and restricted areas need fast local detection without depending fully on human response time.
- Combining motion, distance, heat, and badge authentication reduces false alarms.
- A staged verification pipeline is safer than directly acting on a single sensor trigger.

1. Motivation

Why This Project & Main Objectives

The project integrates sector-based sensing, hardware-PWM servo positioning, multi-bus communication, and interrupt-driven control on the ATmega32. The aim is a complete embedded prototype, not an isolated sensor demo.

1. **Three-sector localization:** 180° field split into left, center, and right sectors using hardware interrupts.
2. **Pan-tilt aiming:** 2-axis servo platform slews to the predefined angle of the triggered sector within 500 ms.
3. **Multi-modal verification:** thermal ΔT using MLX90614 and distance R using VL53L0X over I²C.
4. **Wireless IFF handshake:** application-layer RF badge challenge-response over SPI using nRF24L01+.
5. **Warning actuator:** MOSFET-driven strobe/alarm output after failed authorization and positive sensor verification.
6. **Stretch features:** UART monitoring dashboard and optional MicroSD incident logging.

2. Equipment

Hardware Components & Justification

Purpose	Exact Module Used	Justification
System Controller	ATmega32-16PU	On-chip TWI, SPI, USART, timers, interrupts, and GPIO support the complete prototype without external bus bridges.
Sector Detection	3 × PIR Motion Sensors	Each sector sensor gives a digital edge that can be connected directly to INT0, INT1, or INT2 for low-latency detection.
Thermal Verification	MLX90614 IR Sensor	Factory-calibrated non-contact temperature reading over I ² C; useful for warm-object verification.
Distance Measurement	VL53L0X Laser ToF	Digital laser time-of-flight ranging over I ² C; less affected by air temperature and wind than ultrasonic ranging.
Wireless IFF Link	nRF24L01+ Pair	Low-latency 2.4 GHz packet radio over SPI for challenge-response badge authentication.
Black-Box Logging	MicroSD Card Module	Optional high-capacity non-volatile storage over SPI for incident event logs.
Pan-Tilt Actuation	MG996R Servos (× 2)	Standard hobby servo interface; driven by Timer1 50 Hz PWM outputs OC1A and OC1B.
Telemetry Bridge	FTDI FT232RL	Bridges ATmega32 USART to a PC USB port for the optional monitoring dashboard.
Warning Output	MOSFET + Strobe/Alarm	Low-side switch driven by GPIO to safely control a higher-current warning actuator.
Power Regulation	Dual LM2596 Converters	Separates noisy servo supply from the 5 V logic rail to reduce brown-out risk.
Level Compatibility	3.3 V Regulator / Level Shifter	Protects 3.3 V modules such as VL53L0X and nRF24L01+ when used with a 5 V ATmega32 system.

2. Equipment

Comparison with Simpler Alternatives

Subsystem	Selected Module	Simpler Alternative	Why We Chose Selected Over Alternative
Distance	VL53L0X Laser ToF	HC-SR04 Ultrasonic	Laser ToF gives compact digital ranging and is less affected by air temperature/wind than sound-based ranging.
Thermal	MLX90614 IR Sensor	LM35 Temperature Sensor	LM35 requires contact; MLX90614 supports non-contact warm-target verification.
Wireless	nRF24L01+ RF	HC-05 Bluetooth Module	nRF24L01+ supports fast packet-based custom challenge-response logic without Bluetooth pairing delay.
Logging	MicroSD Card	SPI EEPROM	MicroSD provides much larger storage for optional incident logs.
Controller	ATmega32-16PU	Arduino Uno Board	ATmega32 gives direct DIP-40 pin access and encourages register-level peripheral design.

2. Equipment

Software & Development Tools

Toolchain

- **Microchip Studio / AVR-GCC** – C development, register-level drivers, compilation
- **USBasp / eXtreme Burner** – firmware flashing
- **Proteus VSM** – circuit simulation before fabrication

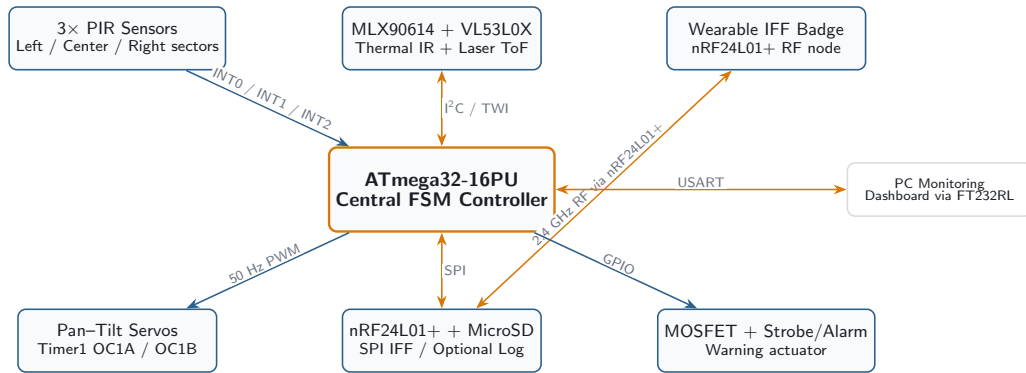
The ATmega32 performs sensing, verification, aiming, and warning control autonomously. The PC dashboard is only a visualization and debugging tool.

Host-Side Tools

- **Python 3 (PySerial)** – UART telemetry parsing
- **Pygame / Tkinter** – radar-style monitoring dashboard

3. System Design

Corrected Block Diagram



→ one-way control/signal path ↔ bidirectional/shared communication path. HC-SR04, LCD, and battery ADC monitoring are intentionally removed for consistency.

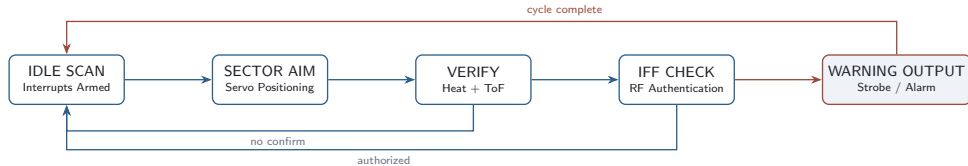
3. System Design

ATmega32 Interface Pinout Allocation Map

Pin(s)	Function	Connected To
PD0 (RXD), PD1 (TXD)	Hardware USART	FTDI FT232RL, 115200 baud
PD2 (INT0)	External interrupt 0	Left sector PIR trigger
PD3 (INT1)	External interrupt 1	Center sector PIR trigger
PB2 (INT2)	External interrupt 2	Right sector PIR trigger
PD5 (OC1A)	Timer1 hardware PWM	Pan servo control signal
PD4 (OC1B)	Timer1 hardware PWM	Tilt servo control signal
PC0 (SCL), PC1 (SDA)	Hardware TWI (I ² C)	MLX90614 and VL53L0X, 7-bit addresses 0x5A and 0x29
PB5/PB6/PB7	Hardware SPI	MOSI / MISO / SCK for nRF24L01+ and MicroSD
PB4 (SS)	SPI master safety pin	Configure as output and keep HIGH
PA1	GPIO output	MicroSD chip select (CS_SD)
PA2	GPIO output	nRF24L01+ CSN
PA3	GPIO output	nRF24L01+ CE
PA4	GPIO input, optional	nRF24L01+ IRQ
PD6	GPIO output	Warning strobe/alarm MOSFET gate

4. Role of the ATmega32

Finite State Machine Logic



- The FSM proceeds through sector detection, servo aiming, thermal/range verification, IFF check, and warning output.
- Feedback paths return the controller to **IDLE SCAN** when the target is not confirmed, the IFF is authorized, or the warning cycle completes.

4. Role of the ATmega32

Peripheral Utilization Breakdown

- **GPIO:** warning MOSFET control, status indicators, SPI chip-select/control lines.
- **External interrupts (INT0/1/2):** low-latency sector detection from three PIR sensors.
- **Timer1 (16-bit):** exact 50 Hz servo PWM using OC1A and OC1B for pan and tilt.
- **Timer0:** 1 ms system tick for non-blocking task scheduling and state-machine timing.
- **Timer2:** spare timer for optional timeout generation or future buzzer output.
- **Hardware TWI (I²C):** 100 kHz bus to MLX90614 and VL53L0X.
- **Hardware SPI:** master mode, shared by nRF24L01+ and optional MicroSD using separate CS/CSN lines.
- **Hardware USART:** 115200 baud telemetry frames, e.g. \$SECTOR,DIST,TEMP,STATUS#.

5. Implementation Plan

Milestones

Project Update 1 (Wks 1–3)

- 1.1 Bare-metal clock, GPIO, and register setup
- 1.2 Timer1 50 Hz PWM for pan-tilt servos
- 1.3 Hardware I²C driver: MLX90614 + VL53L0X
- 1.4 External interrupts for three-sector PIR detection

Deliverable: turret aims to activated sector and sends distance/temperature over UART.

Final Demonstration (Wks 4–6)

- 2.1 SPI driver for nRF24L01+ and optional MicroSD logging
- 2.2 Application-layer IFF challenge-response handshake
- 2.3 PC monitoring dashboard over UART
- 2.4 MOSFET-driven strobe/alarm warning output
- 2.5 Power isolation, level shifting, enclosure, and stress testing

Deliverable: sector detection, servo aiming, thermal/range verification, IFF, warning output, and dashboard.

5. Implementation Plan

Group Member Responsibilities

Member 1: Firmware & Control

- Register-level drivers: GPIO, Timer1 PWM, Timer0 tick, I²C, external interrupts
- Core FSM architecture
- Timing-loop optimization

Member 2: Hardware, Power & Actuation

- Pan-tilt turret assembly
- 5 V/6 V power distribution and decoupling
- Sensor calibration, level shifting, MOSFET warning circuit

Member 3: Wireless, Storage & GUI

- SPI management for nRF24L01+ and optional MicroSD
- IFF challenge-response protocol logic
- Python monitoring dashboard

6. Use of Other Controllers

Required Auxiliary IFF Badge

Main System

The ATmega32 is the sole main controller. It handles the fixed station: PIR detection, servo aiming, sensor verification, RF challenge initiation, optional logging, UART dashboard, and warning output.

Auxiliary Controller – Wearable IFF Badge

- **Device:** Arduino Nano / ATtiny85, battery-powered and worn by authorized personnel.
- **Why required:** the badge is physically separate from the fixed turret station, so it needs its own small controller.
- **Task:** receives a 2.4 GHz challenge packet and returns a keyed response using its own nRF24L01+ module.
- **Interface:** wireless RF only; it performs no main-system sensing, aiming, logging, or warning-output tasks.

7. Testing & Success Criteria

Measurable Criteria & Test Methodology

Metric	Target	Test Method	Priority
Sector detection latency	< 100 ms	PIR trigger to MCU state change	Essential
Turret slew speed	< 500 ms	Trigger-to-sector-angle timing	Essential
Servo PWM stability	50 Hz on OC1A/OC1B	Oscilloscope waveform check	Essential
Distance verification	± 30 mm, 10–120 cm	VL53L0X indoor matte-target bench test	Essential
Thermal verification	$\Delta T \geq 3^{\circ}\text{C}$ above background	Controlled warm-object test	Essential
I ² C bus stability	Clean SCL @ 100 kHz	Multi-device signal-integrity check	Essential
IFF handshake execution	< 50 ms	RF challenge-response timing	Essential
Warning output	Correct MOSFET switching	Strobe/alarm activation test	Essential
Telemetry streaming rate	≥ 10 Hz, no frame loss	PC dashboard continuous update	Optional
MicroSD black-box log	Event timestamp/sector/heat/range	File readback verification	Optional

Essential features: PIR sector detection, Timer1 servo aiming, I²C heat/range verification, IFF handshake, and warning output. Optional features: MicroSD logging and PC monitoring dashboard.

8. Challenges & Future Scope

Anticipated Issues & Scaling Path

Technical & Hardware Challenges

- **SPI sharing:** nRF24L01+ needs low-latency packets while MicroSD performs longer writes → separate CS lines and bus-release routines.
- **3.3 V modules:** VL53L0X and nRF24L01+ require correct regulation/level compatibility → level shifting or safe breakout boards.
- **Servo noise:** high current spikes can reset the MCU → isolated servo rail, common ground, and bulk decoupling.
- **Sensor certainty:** MLX90614 confirms a warm object, not identity → combine heat, range, sector, and IFF results.

Future Scope

- **Computer vision:** ESP32-CAM co-processor for visual target classification.
- **360° coverage:** stepper/continuous-rotation mechanism with slip ring support.
- **Long-range communication:** LoRa telemetry network for multi-unit perimeter monitoring.